MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED METALLIZED PLASTIC DIELETHIC, DC, IN NONMETAL CASES, ESTABLISHED RELIABILITY

STYLES CFR13 AND CFR14

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements of acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-55514.

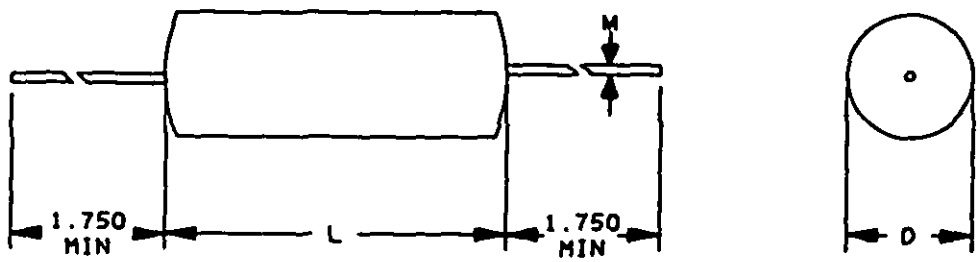
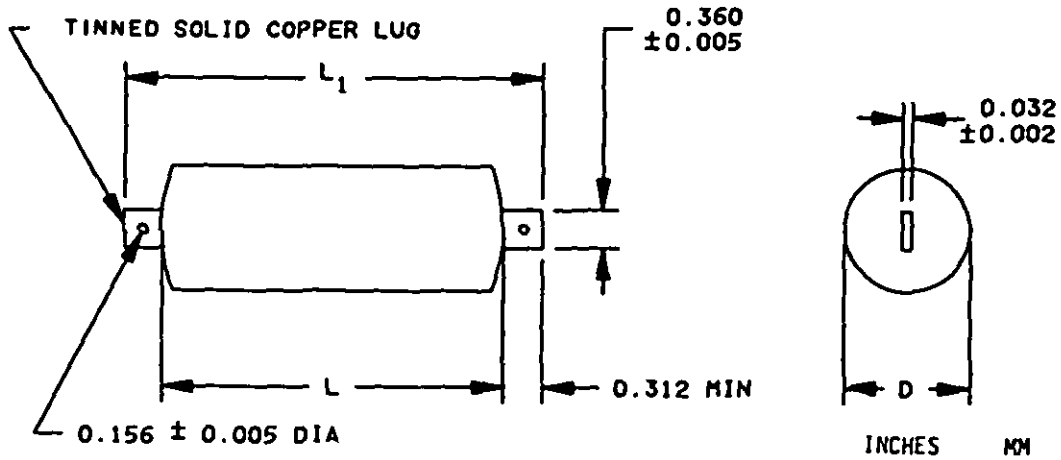


FIGURE 1. Style CFR13 capacitors.



INCHES	MM
.002	.05
.005	.13
.032	.81
.156	3.96
.312	7.93
.360	9.14
1.750	44.45

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 2. Style CFR14 capacitors.

FSC 5910

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:29Z
Check the source to verify that this is the current version before use.

TABLE I. Capacitor characteristics.

Characteristic L, style CFR13 capacitors														
Type designation 1/	Capacitance nominal	DC rated voltage (at 85°C)	Capacitance tolerance available	D	L ±.062 ±(1.57)	M 2/	ESR Ohms 20-100 kHz Max	Maximum ripple current (AMPS RMS) 20-100 kHz case temperature						
								+25°C	+35°C	+45°C	+55°C	+65°C	+75°C	+85°C
CFR13ALB105--	1.0	100	J, K, M	.469 (11.91) ±.062 (1.57)	.750 (19.05)	.032 (.81)	.015	9.2	8.5	7.8	7.0	6.0	4.9	4.5
CFR13ALB205--	2.0	"	"	.534 (13.56) ±.062 (1.57)	.938 (23.81)	.032 (.81)	.012	10.8	10.0	9.1	8.2	7.0	5.8	5.3
CFR13ALB305--	3.0	"	"	.624 (15.85) ±.093 (2.36)	.938 (23.81)	.040 (1.02)	.011	12.1	11.2	10.3	9.2	8.0	6.5	5.9
CFR13ALB505--	5.0	"	"	.640 (16.26) ±.093 (2.36)	1.250 (31.75)	.040 (1.02)	.010	13.8	12.7	11.6	10.4	9.0	7.4	6.7
CFR13ALB106--	10.0	"	"	.805 (20.45) ±.093 (2.36)	1.500 (38.10)	.040 (1.02)	.009	15.0	15.0	14.2	12.7	11.0	9.0	8.2
CFR13ALB206--	20.0	"	"	.875 (22.23) ±.125 (3.18)	2.250 (57.15)	.040 (1.02)	.008	15.0	15.0	15.0	15.0	13.6	11.1	10.0
CFR13ALB306--	30.0	"	"	1.075 (27.31) ±.125 (3.18)	2.250 (57.15)	.040 (1.02)	.006	15.0	15.0	15.0	15.0	15.0	12.4	11.4
CFR13ALC105--	1.0	200	"	.450 (11.43) ±.062 (1.57)	1.250 (31.75)	.032 (.81)	.020	7.3	7.3	7.3	7.3	7.2	5.9	5.4
CFR13ALC205--	2.0	"	"	.605 (15.37) ±.093 (2.36)	1.250 (31.75)	.032 (.81)	.015	12.0	12.0	11.3	10.1	8.7	7.1	6.5
CFR13ALC305--	3.0	"	"	.654 (16.61) ±.093 (2.36)	1.500 (38.10)	.040 (1.02)	.013	15.0	13.8	12.6	11.3	9.8	8.0	7.3
CFR13ALC505--	5.0	"	"	.769 (19.53) ±.093 (2.36)	1.750 (44.45)	.040 (1.02)	.011	15.0	15.0	14.7	13.1	11.4	9.3	8.5
CFR13ALC106--	10.0	"	"	.905 (22.99) ±.125 (3.18)	2.250 (57.15)	.040 (1.02)	.009	15.0	15.0	15.0	15.0	13.8	11.3	10.3
CFR13ALC206--	20.0	"	"	1.315 (33.40) ±.125 (3.18)	2.250 (57.15)	.040 (1.02)	.006	15.0	15.0	15.0	15.0	15.0	14.1	12.8
CFR13ALE105--	1.0	400	"	.620 (15.75) ±.093 (2.36)	1.500 (38.10)	.040 (1.02)	.019	9.5	9.5	9.5	9.5	9.5	7.8	7.1
CFR13ALE205--	2.0	"	"	.802 (20.37) ±.093 (2.36)	1.750 (44.45)	.040 (1.02)	.015	15.0	15.0	15.0	13.4	11.6	9.5	8.7
CFR13ALE305--	3.0	"	"	.961 (24.41) ±.125 (3.18)	1.750 (44.45)	.040 (1.02)	.012	15.0	15.0	15.0	15.0	13.1	10.7	9.8
CFR13ALE505--	5.0	"	"	1.067 (27.10) ±.125 (3.18)	2.250 (57.15)	.040 (1.02)	.010	15.0	15.0	15.0	15.0	15.0	12.5	11.4
CFR13ALE106--	10.0	"	"	1.543 (39.19) ±.125 (3.18)	2.250 (57.15)	.040 (1.02)	.006	15.0	15.0	15.0	15.0	15.0	15.0	14.1

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:29Z
Check the source to verify that this is the current version before use.

TABLE I. Capacitor characteristics - Continued.

Characteristic L, style CFR14 capacitors														
Type designation 1/	Capacitance nominal	DC rated voltage (at 85°C)	Capacitance tolerance available	D	L ±.078 ±(1.98)	L1 ±.062 ±(1.57)	ESR Ohms 20-100 kHz Max	Maximum ripple current (AMPS RMS) 20-100 kHz case temperature						
								+25°C	+35°C	+45°C	+55°C	+65°C	+75°C	+85°C
CFR14LLB105--	1.0	100	J, K, M	.469 (11.91) ±.062 (1.57)	.922(23.42)	1.640(41.66)	.015	10.3	9.5	8.7	7.8	6.7	5.5	5.0
CFR14LLB205--	2.0	"	"	.534 (13.56) ±.062 (1.57)	1.110(28.19)	1.828(46.43)	.012	12.0	11.0	10.0	8.9	7.8	6.3	5.8
CFR14LLB305--	3.0	"	"	.624 (15.85) ±.093 (2.36)	1.110(28.19)	1.828(46.43)	.011	13.3	12.3	11.2	10.0	8.7	7.1	6.5
CFR14LLB505--	5.0	"	"	.640 (16.26) ±.093 (2.36)	1.422(36.12)	2.140(54.36)	.010	14.8	13.7	12.5	11.2	9.7	7.9	7.2
CFR14LLB106--	10.0	"	"	.805 (20.45) ±.093 (2.36)	1.672(42.47)	2.390(60.71)	.009	17.8	16.5	15.0	13.5	11.7	9.5	8.7
CFR14LLB206--	20.0	"	"	.875 (22.23) ±.125 (3.18)	2.422(61.52)	3.140(79.76)	.008	21.6	20.0	18.3	16.4	14.2	11.6	10.6
CFR14LLB306--	30.0	"	"	1.075 (27.31) ±.125 (3.18)	2.422(61.52)	3.140(79.76)	.006	24.3	22.5	20.5	18.4	15.9	13.0	11.9
CFR14LLC105--	1.0	200	"	.450 (11.43) ±.062 (1.57)	1.422(36.12)	2.140(54.36)	.020	7.3	7.3	7.3	7.3	7.3	6.4	5.8
CFR14LLC205--	2.0	"	"	.605 (15.37) ±.093 (2.36)	1.422(36.12)	2.140(54.36)	.015	14.3	13.3	12.1	10.8	9.4	7.7	7.0
CFR14LLC305--	3.0	"	"	.654 (16.61) ±.093 (2.36)	1.672(42.47)	2.390(60.71)	.013	15.9	14.7	13.5	12.0	10.4	8.5	7.8
CFR14LLC505--	5.0	"	"	.769 (19.53) ±.093 (2.36)	1.922(48.82)	2.640(67.06)	.011	18.3	17.0	15.5	13.9	12.0	9.8	8.9
CFR14LLC106--	10.0	"	"	.905 (22.99) ±.125 (3.18)	2.422(61.52)	3.140(79.76)	.009	22.4	20.7	18.9	16.9	14.6	12.0	10.9
CFR14LLC206--	20.0	"	"	1.315 (33.40) ±.125 (3.18)	2.422(61.52)	3.140(79.76)	.006	27.4	25.4	23.2	20.7	17.9	14.7	13.4
CFR14LLE105--	1.0	400	"	.620 (15.75) ±.093 (2.36)	1.672(42.47)	2.390(60.71)	.019	9.5	9.5	9.5	9.5	9.5	8.3	7.5
CFR14LLE205--	2.0	"	"	.802 (20.37) ±.093 (2.36)	1.922(48.82)	2.640(67.06)	.015	15.0	15.0	15.0	14.2	12.3	10.0	9.1
CFR14LLE305--	3.0	"	"	.961 (24.41) ±.125 (3.18)	1.922(48.82)	2.640(67.06)	.012	21.1	19.5	17.8	15.9	13.8	11.3	10.3
CFR14LLE505--	5.0	"	"	1.067 (27.10) ±.125 (3.18)	2.422(61.52)	3.140(79.76)	.010	24.4	22.6	20.6	18.5	16.0	13.1	11.9
CFR14LLE106--	10.0	"	"	1.543 (39.19) ±.125 (3.18)	2.422(61.52)	3.140(79.76)	.006	30.0	27.8	25.4	22.7	19.7	16.1	14.7

1/ The complete type designation will include additional symbols to indicate capacitance tolerance and failure rate level (M, P, R, and S).

2/ Comparable AWG sizes are No. 20 AWG (.032) and No. 18 AWG (.040).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:29Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figures I and II and table I.

Case: Non-hermetic.

Dielectric: Polypropylene, metallized.

Capacitance (Cap.) value: See table I.

Cap. tolerance: See table I.

Rated voltage: See table I.

Operating temperature range: -55°C to 105°C for dc operation.

Failure rate (FR) level: M, P, R or S.

Preconditioning: Not applicable.

Burn-in: In accordance with MIL-C-55514 except paragraph (b) shall be +105° +0, -5°C, and the applied voltage shall be 120 percent of the dc voltage specified in table I.

Dielectric withstanding voltage (DWV): In accordance with MIL-C-55514 except charge time 2 minutes maximum at 200 percent rated dc voltage.

Insulation resistance: In accordance with MIL-C-55514 except test voltage to be 100 V dc.

Dissipation factor (DF): Not more than 0.1 percent when measured in accordance with MIL-C-55514.

Low temperature conditioning and capacitance change with temperature: In accordance with MIL-C-55514.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.

ΔCap: Maximum of +0.25 percent.

Life (at 105°C): In accordance with MIL-C-55514.

Accelerated condition: 140 percent of 105°C rated voltage.

Rated condition: 100 percent of +85°C rated voltage.

DF: .1 percent maximum.

IR: (terminal to terminal)

(at +25°C): Not less than 50.0 percent of initial requirement.

ΔCap: Maximum of 1.0 percent.

DF: (at +25°C) after life : .12 percent maximum.

AC conditioning: Not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:29Z
Check the source to verify that this is the current version before use.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Navy - OS
Air Force - 17, 99
DLA - ES

User activities:
Navy - AS, MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5910-1427-3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:29Z
Check the source to verify that this is the current version before use.